DIRECTORATE OF GOVERNMENT EXAMINATIONS, CHENNAI – 6
HIGHER SECONDRAY SECOND YEAR EXAMINATION- MARCH -2020
CHEMISTRY KEY ANSWER (NEW SYLLABUS)

NOTE:

1. Answers written with **Blue or Black** ink only to be evaluated.
2. Choose the correct answer and write with option code.
3. If one of them (option or answer) is wrong, then award Zero mark only.

MAXIMUM MARKS : 70

PART – I **15 x 1 = 15**

TYPE-A			TYPE-B		
Q. No	Option	Answers	Q. No	Option	Answers
1	b	(1)-(ii), (2)-(i), (3)-(iv), (4)-(iii)	1	b	$H_2N_2O_2$
2	a	Electromagnetic separation	2	c	o-phenol sulphonic acid
3	d	Sc	3	b	5F
4	c	Therapeutic index	4	d	32%
5	c	basic, acidic, basic	5	c	Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
6	b	TACGAACT	6	b	(1)-(ii), (2)-(i), (3)-(iv), (4)-(iii)
7	c	2,4-dimethyl aniline	7	a	Electromagnetic separation
8	b	5F	8	c	2,4-dimethyl aniline
9	c	Both Assertion and Reason are true but Reason is not the correct explanation of Assertion	9	c	SN^2 reaction
10	d	Lithium-ion battery	10	c	Therapeutic index
11	b	$H_2N_2O_2$	11	d	Lithium-ion battery
12	c	SN^2 reaction	12	d	Sc
13	d	32%	13	a	half-life period
14	a	half-life period	14	c	basic, acidic, basic
15	c	o-phenol sulphonic acid	15	b	TACGAACT

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PART – II

Q.No. 24 is compulsory **2 x 6 = 12**

16	Bleaching powder Correct equation Chlorine + calcium hydroxide (or) formula or name only (or) any correct explanation	2 1	2
17	(i) Tungsten- d block (ii) Ruthenium-d-block (iii) Promethium-f-block (iv) Einstenium-f-block (v)	4 x ½	2
18	(i) $[Cr(H_2O)_6]Cl_3$ (ii) $[Cr(H_2O)_5Cl]Cl_2.H_2O$ (iii) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ (Any two)	2 x 1	2
19	Octahedral void is **6** Tetrahedral void is **12**	2 x 1	2
20	Lewis acid : accept a pair of electron + any one example Lewis base: donate a pair of electron + any one example (or) any correct explanation + any one example	1 1 1+1	2
21	Dispersed phase : liquid Dispersion medium :solid	2 x 1	2
22	$Pd/BaSO_4$ $BaSO_4$ acts as catalytic poison to pd and prevents further reduction of aldehyde	1 ½ + ½	2
23	Correct Equation **(or)** not mentioning NaoH **(or)** Explanation only	2 1 ½ 1	2
24	due to the repulsive interaction between the two bulkier alkyl groups (or) due to steric effect (or) interaction	2	2

PART – III

Q.No. 33 is compulsory **3 x 6 = 18**

25	Chromyl chloride test Balanced equation (only main equation) Unbalanced equation (or) Statement : (or) salt + Potassium dichromate + conc sulphuric acid gives red orange vapours	3 2 2 2	3

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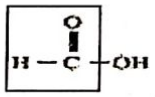
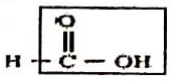
26	<u>$[Sc(H_2O)_6]^{3+}$ colourless</u>: (i) The Outer electronic configuration of central metal ion Sc^{3+} is $3d^0$ (ii) Due to presence of vacant or empty d-orbital (iii) d-d transition is not possible in this complex. So it is colourless.	1 1 1	3
27	Henderson equation $[H_3O^+] = K_a \frac{[acid]_{eq}}{[base]_{eq}}$ Or correct explanation for the above equation $-\log [H_3O^+] = -\log K_a - \log \frac{[acid]}{[salt]}$ $pH = pK_a + \log \frac{[salt]}{[acid]}$	1 1 1	3
28	<u>Cathodic protection method:</u> (i) By using metals such as Mg or Zn which is corroded more easily than iron can be used as sacrificial anode and the iron material act as cathode. (ii) Iron is protected, but Mg or Zn is corroded. (OR) Mere mentioning Mg, Zn	2 1 1	3
29	<u>Shapes of colloids:</u> (i) Spherical (ii) Disc or Plate like (iii) Rod like	1 1 1	3
30	(i) Formic acid contains both an aldehyde as well as an acid group, So formic acid reduces Tollens reagent. (ii) Acetic acid does not consist of aldehyde group, so it does not reduces Tollens reagent. (OR)  Aldehyde group  Carboxylic acid group	2 1 2	3
31	fibrous protein and globular proteins. <u>fibrous protein</u> linear molecules (or) insoluble in water (or) Example: Keratin (or)Collagen (or) any relevant explanation <u>globular proteins.</u> spherical shape (or) soluble in water (or) example : enzymes (or) myoglobin (or) Insulin (or) any relevant explanation.	1 1 1	3
32	<u>Advantages of food additives:</u>	3 x 1	3

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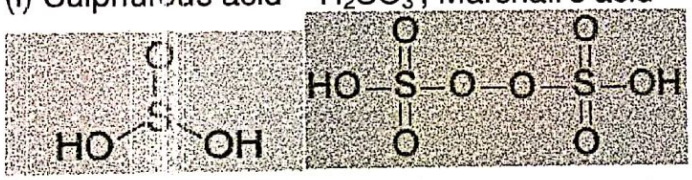
	Any three advantages		
33	due to the presence of inner d and f-electrons which has poor shielding effect compared to s and p-electrons. (or) Any other relevant explanation	3	3

PART – IV

5 x 5 = 25

34 a	Zone refining fractional crystallization impurities will prefer to be in the molten region. Explanation Example Si (or) Ge(or) Ga (or semiconductors	 1 1 2 1	5
(b)	(i)<u>Conditions for catenation:</u> Any two conditions. (ii) HF reacts with glass balanced equation (or) Any unbalanced equation (or) explanation	 2 1 2 1	5
35 (a)	(i) Sulphurous acid – H_2SO_3 , Marshall's acid - $H_2S_2O_8$  (ii)<u>IUPAC Name:</u> (A) diamminesilver(I)ion (B) pentaamminechlorocobalt(III)ion (Hint: the ligand name "chloro" can also be written as "chlorido")	½ + ½ ½ + ½ 2 x 1 ½	5
(b)	(i) * Magnetic property: Unpaired electrons=4, or Paramagnetic Magnetic moment: $\mu_s = \sqrt{n(n+2)} = 4.899\,BM$ (ii) Frenkel defect: dislocation of ions / interstitial position. Explanation picture (or) example	 1 ½ + ½ 1 1 1	5

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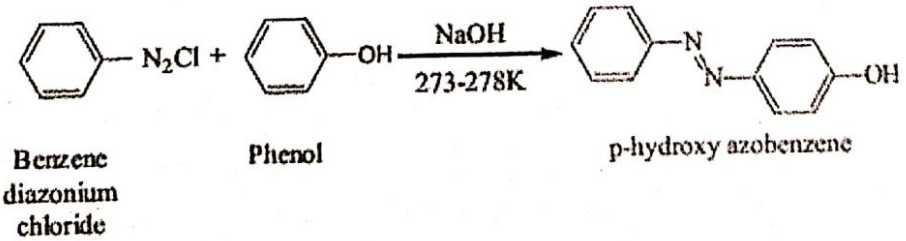
36 (a)	Rate=k[A]^1----------1 $-\frac{d[A]}{dt} = k[A]^1$ $-\frac{d[A]}{[A]} = kdt$ ----------1 $\int_{[A_0]}^{[A]} \frac{-d[A]}{[A]} = k\int_0^t dt$ ----------1 $\ln\left(\frac{[A_0]}{[A]}\right) = kt$ ----------1 $k = \frac{2.303}{t}\log\left(\frac{[A_0]}{[A]}\right)$ ----------1 (or) Any correct equations	5	5
(b)	(i)pH value: (A) Vinegar – 2, or (acidic) (B) Black coffee – 5 or (acidic) (C) Baking soda – 9, or (basic) (D) Soapy water – 12 or (basic) (ii) $\kappa = \frac{1}{R}\left(\frac{l}{A}\right)$ | $l = 1.5\ cm = 1.5\times10^{-2} m$ $\kappa = \frac{1}{15\Omega}\times\frac{1.5\times10^{-2} m}{4.5\times10^{-4} m^2}$ | $A = 4.5\ cm^2 = 4.5\times(10^{-4}) m^2$ $= 2.22\ Sm^{-1}$ | $R = 15\Omega$	4 x ½ 3 x 1	5
37 (a)	(i) Differences between chemisorption and physisorption: Any three differences (ii) Vulcanization Correct definition (or) Natural rubber and sulphur (mentioning these two only)	3 x 1 2 1	5
(b)	(i) Coupling reaction of phenol: Correct equation  not mentioning NaoH (or) Explanation	2 1 ½ 1	5

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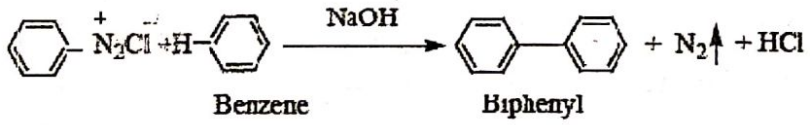
	(ii) <u>Preparation using Grignard reagent</u>: (A) Correct equation using H-CHO and C_2H_5-MgBr (OR) Any other correct equation (Grignard reagent must) (B) Correct equation using CH_3-CHO and CH_3-MgBr (OR) Any other correct equation (Grignard reagent must) (OR) Explanation only	$1^1/_2$ $1^1/_2$ ½ + ½	
38 (a)	(i) <u>Formalin:</u> 40% aqueous solution of formaldehyde is called formalin. Use : Any one use (ii) <u>Glycosidic linkage:</u> In disaccharides or (Oligo) or (Poly saccharides) two monosaccharide's are linked by oxide linkage called glycosidic linkage.	1 1 3	5
(b)	(i) <u>Gomberg reaction:</u>Correct equation  (OR) not mentioning NaOH - (or) Explanation only (ii) A – CH_3CN (or) Methyl cyanide (or) ethanenitrile B – CH_3NC (or) Methyl isocyanide.	3 1 ½ 1 1 1	5

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Directorate of Government Examinations, Chennai-6

Higher Secondary second year Re Examination 2020

Chemistry Key Answer (New Syllabus)

Note:-

1. **Answer written with Blue or Black ink only to be evaluated**
2. **Choose the correct answer and write with option code**

Max. Marks – 70

PART – I **15x1=15**

Type – A			Type - B	
Q.No	Option	Answer	Q.No	Answer
1	d	C_2 carbon	d	Strongly acidic
2	d	Sandmeyer reaction	c	Bayer's reagent
3	c	$O_{2(g)}+2H_2O_{(l)}+4^{e-} \rightarrow 4OH^-(aq)$	a	$[Cu(NH_3)_4]^{2+}$
4	B	acetone	c	$\rho = \frac{nM}{a^3N_A}$
5	D	Strongly acidic	d	C_2 carbon
6	B	$B_3N_3H_6$	b	+7
7	A	Metal disappointment with zinc	b	$E + S \leftrightharpoons ES \rightarrow P+E$
8	C	Bayer's reagent	c	$O_{2(g)}+2H_2O_{(l)}+4^{e-} \rightarrow 4OH^-(aq)$
9	D	Lidocaine	c	(1)-iii; (2)-(iv); (3)-(i); (4)- (ii)
10	C	$\rho = \frac{nM}{a^3N_A}$	b	acetone
11	B	+7	b	$B_3N_3H_6$
12	A	Pyrogallol	a	Metal disappointment with zinc
13	C	(1)-iii; (2)-(iv); (3)-(i); (4)- (ii)	d	Lidocaine
14	A	$[Cu(NH_3)_4]^{2+}$	a	Pyrogallol
15	B	$E + S \leftrightharpoons ES \rightarrow P+E$	d	Sandmeyer reaction

PART – II

Q.No. 24 is compulsory **2 x 6 = 12**

16	Used as a flux		2
17	Double salt It loose their identity And dissociates into Their constituent simple Ions (OR) dissociate into their constituent simple ion Coordination compound it does not loose its identity and never dissociate to give simple ions never dissociate to simple ions		2 1
18	Correct definition (OR) resist P^H change		2 1
19	Correct definition		2
20	Scattering of light by sol particles (OR) Any correct explanation		2
21	Hexamethylene tetramine (OR) $(CH_2)\ N_4$ $6HCHO+4NH \rightarrow (CH_2)_6N_4+6H_2O$ (OR)correct explanation (or) unbalanced equation	1 1 1/2	2
22	Aniline donates its lone pair to the lewis acid $AlCl_3$ to form an adduct (OR) any correct explanation		2
23	a)vitamin D b) vitamin C	1 1	2
24	A - LiH (OR) Lithium hydride B - B_2H_6 (OR) Diborane C - $LiBH_4$ (OR) Lithium Boro Hydride	1/2 ½ 1	2

PART – III

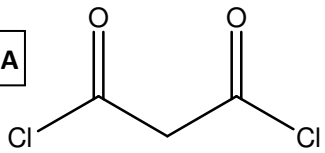
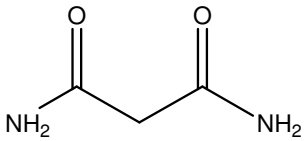
Q.No. 33 is compulsory **3 x 6 = 18**

25	Sulphide ores Ex:- 1.Galena or Pbs 2. Zinc blende or Zns (or) any other examples	1 1 1	3
26	a) BrF – Sp^3 b) BrF_5 -- Sp^3d^2 c) BrF_3 -- Sp^3d	1 1 1	3
27	Correct diagram Without mentioning energy (or) Δo or d-orbitals Nearly correct diagram	 2 1	3
28	Any three differences	3x1=3	3

29	Oswald dilution law $CH_3COOH \leftrightharpoons H^+ + CH_3COO^-$ $K_a = \frac{[H^+][CH_3COO]}{[CH_3COOH]}$ $K_a = \frac{(\propto.C)(\propto.C)}{(1-\propto)C}$ $K_a = \frac{\propto^2 C}{1-\propto}$ (or) $[H^+] = \sqrt{K_a.C}$ (or) $\propto = \sqrt{K_a/C}$	 ½ 1 ½ 1	3
30	a) – SN^2 mechanism b) – SN^1 mechanism	1½ 1½	3
31	Any correct 3 tests	3x1=3	3
32	Three steps Without mentioning Alcoholic KOH (or) R-X (or) aq KOH Or Mere statement	3x1=3 2 1	3
33	When the $CaCO_3$ particles are smaller so **surface area** for the acid is high	3	3

PART – IV 5X5=25

34a) (i)	Preparation of potash alum (i) Correct equation (ii) Unbalanced equation (or) (iii) Mere equation	 $2x1\frac{1}{2}=3$ 2x1 = 2 1	3
(II)	A – Optical Isomerism B – Geometrical Isomerism (or) Cis- trans isomerism	1 1	2
b) (i)	Deacon's process Correct equation (or) Unbalanced equation (or) Mere explanation	 2 1 1	2
(ii)	Sulphuric acid is a dibasic acid Correct equation (or) Unbalanced equation (or) Mere explanation	 2x11/2 = 3 2x1 = 2 1	3
35(a)	Correct explanation for lanthanide contraction Consequences	2 3x1 = 3	5
(b)(i)	Co-ordination number – 3 Structure - Trigonal planer	1 1	2
(ii)	$CH_3COCl > CH_3COOCOCH_3 > CH_3COOC_2H_5 > CH_3CONH_2$ Reason ; (i) Basic nature of leaving group (ii) Resonance	1 1 1	3

36.(a) (i)	Rate constant K = $1.54 \times 10^{-3}\ s^{-1}$ $t_{½} = \frac{0.6931}{k_1}$ $= \frac{0.6931}{1.54 \times 10^{-3}}$ = 450 seconds (1/2 +1/2)	1 1 1	3
(ii)	(A) $HS^-_{aq} + HF \rightleftharpoons F^- + H_2S$ HS^- Base 2 HF – Acid 1 F^- - Base1 H_2S - Acid 2 (B) $HPO_4^{2-} + SO_3^{2-} \rightleftharpoons PO_4^{3-} + HSO_3^-$ HPO_4^{2-} - Acid 1 SO_3^{2-} - base 2 PO_4^{3-} - base 1 HSO_3^- - Acid 2	1 1	2
36(b)	Kohlraush's law Correct definition One application with explanation	3 2	5
37(a)	Any 5 Characteristics (or)	5x1 =5	5
(b)	Victor meyer's test (i) Correct equation s for primary alcohol (or) Without reagents (or) Statements only (ii) Correct equations for secondary alcohol (or) With out reagents (or) Statements (iii) Correct equations for tertiary alcohol (or) With out reagents (or) Statements	2 1 1/2 1 11/2 1 11/2 1	5
38(a)	(i) A  B 	1 1	2

	(ii) 1)Ribosomal RNA (or) rRNA any correct explanation 2) Messenger RNA (or) mRNA any correct explanation 3) Transfer RNA (or) tRNA any correct explanation	1/2 1/2 1/2 1/2 1/2 1/2	3
38(b)	(i) Antioxidants Substance which retard the oxidative deteriorations of food. To prevent the oxidation of the fats and oils chemical BHT , BHA are added as food additives (or) Any correct explanation	2 1	3
	(ii) A – Antacid B- Analgesics (or) Antipyritics C- Antimicrobials D- Anaesthetics	1/2 1/2 1/2 1/2	2

1E-06

DEPARTMENT OF GOVERNMENT EXAMINATIONS

Higher Secondary Second Year Common Examinations – September-2020

Key Answers for Chemistry – English medium

PART - I

Q.No	Option	Answer
1	d	In the metallurgy of gold, the metal is leached with dilute Sodium Chloride solution.
2	c	Al
3	d	$Na_2B_4O_7.10H_2O$
4	d	Phosphine
5	d	Group number 3 and Period number 6
6	c	$[Co(NH_3)_3Cl_3]$
7	b	68 %
8	a	First order
9	c	OH^- and F^- respectively
10	a	Adsorption is exothermic
11	b	S_N^2 mechanism
12	a	$(CH_3)_2C(OH)C_6H_5$
13	a	Thorpe nitrile condensation
14	d	Vitamin B_2
15	a	(1) – (iii), (2) – (iv), (3) – (i), (4) – (ii)

PART – II

16	Gangue and Slag: (a) Gangue: Correct explanation + one example (b) Slag: correct explanation + one example	 ½ + ½ ½ + ½	2
17	Uses of Helium: Any two points	2 x 1=2	2

18	Interstitial compounds		2A
	Compound that is formed when small atoms like H,B,C (or) N are trapped in the interstitial holes in a metal lattice. (or)	2	2
	Example only.	1	
19	Distinguish between Isotropy and Anisotropy:		
	Any one correct difference (or)	2	2
	Diagram only (or) Example	1	
20	Order of the reaction:		
	Rate = $k[x]^n[y]^m$ (or) only substitution	1	2
	Order = 1 (or) n + m = 1	1	
21	pH calculation:		
	$pH = 7 + \frac{pK_a}{2} + \frac{\log C}{2}$	1	2
	$= 7 + \frac{4.74}{2} + \frac{\log 0.1}{2}$ pH = 8.87	½ + ½	
22	Glycerol to acrolein:		
	Correct equation with dehydrating agents (or)	2	2
	Correct equation without dehydrating agent (or) Explanation only	1	
23	Denaturation of proteins:		
	The process of losing its higher order structure without losing the primary structure. (or) any correct explanation.(or)	2	2
	Only example	1	

24	Preparation of aryl halide $C_6H_5\text{-}\overset{+}{N_2}\bar{C}l \xrightarrow{Cu_2Cl_2/HCl} C_6H_5\text{-}Cl + N_2$ (or) $C_6H_5\text{-}\overset{+}{N_2}\bar{C}l \xrightarrow{Cu_2Br_2/HBr} C_6H_5\text{-}Br + N_2$ Any one reaction enough (or) Only explanation	2 1	2

PART – III

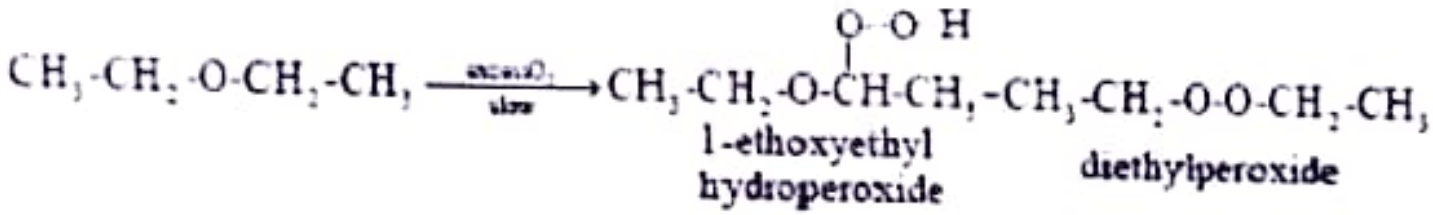
25	Factors responsible for the anomalous behavior of first element of the p-block: (i) Small size (ii) High ionisation enthalpy and high electronegativity (iii) Absence of d-orbitals in their valance shell.	1 1 1	3
26	Metal with +1 oxidation state in 3d-series: (i) Copper (ii) In +1 oxidation state it forms Cu^+ ion with stable $3d^{10}$ Configuration.(OR) It attains stable configuration	1 2 1	3
27	Metal complexes and its metal ions used in biological system: (i) RBC(heme group) - Fe^{2+} (ii) Chlorophyll - Mg^{2+} (iii)Vitamin B_{12}(Cyanocobalamine) - Co^+ (iv) Carboxy peptides -Zn^{2+} (Any three)	3 x 1=3	3
28	Ionic product of water: Correct definition (or) $K_w = [H_3O^+][OH^-]$ $K_w = 1 \times 10^{-14}$	2 1	3
29	Inversion of phase: The change of W/O emulsion into O/W emulsion Correct example	2 1	3

30	Benedict's solution test Correct explanation (or) Correct equation with colour change (or) Cu^{2+} is reduced by aldehyde to give red precipitate of Cu_2O (or) A mixture of $CuSO_4$ + sodium citrate + NaOH	3 2 1	3
31	Biological importance of lipids: Any three points	3 x 1 =3	3
32	Preparation of neoprene: $nCH_2=\underset{Cl}{\underset{\vert}{C}}-CH=CH_2 \xrightarrow[\text{Polymerisation}]{\text{free radical}} \left[CH_2-\underset{Cl}{\underset{\vert}{C}}=CH-CH_2\right]_n$ (or) The free radical polymerization of the monomer, 2-chlorobuta-1,3-diene(chloroprene) gives neoprene.	3 2	3
33	Mass of silver deposited at the cathode: m = ZIt $m = \frac{108\ gmol^{-1}}{96500\ Cmol^{-1}} \times 3600\ C$ m = 4.03 g.	1 1 1	3

PART – IV

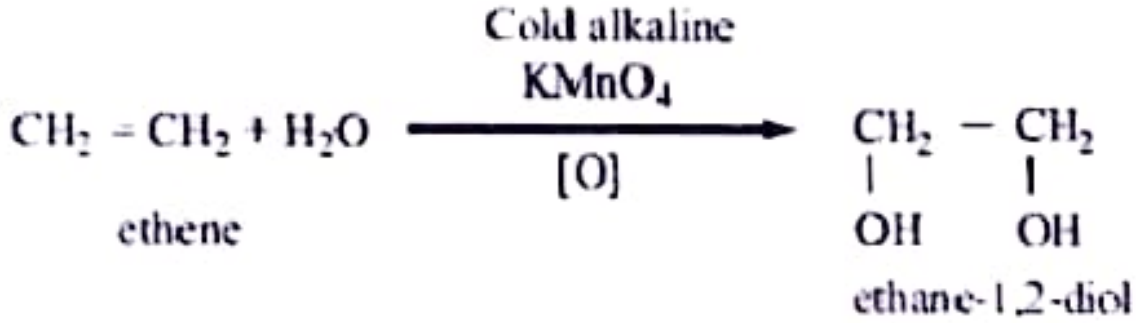
34 (a)	(i) Differences between minerals and ores:		
	Any two points	2 x 1=2	
	(ii) Correct balanced equation		
	$Cl_2 + 2NaOH \longrightarrow NaOCl + NaCl + H_2O$	1½	
	$3Cl_2 + 6NaOH \longrightarrow NaClO_3 + 5NaCl + 3H_2O$	1½	5
	(or)		
	Unbalanced equation	1	
	(or)		
	Correct Explanation only	1	
(b)	(i) catenation:		
	Correct definition	2	
	(ii) Holmes signal:		5
	1. A mixture of calcium carbide and calcium phosphide 2. Liberates phosphine and acetylene 3. Phosphine catches fire and ignites acetylene.	3x1=3	
35 (a)	Postulates of Werner's theory:		
	Five postulates	5x1=5	5
(b)	(i) Schottky defect:		
	Correct explanation (two points) Example + Diagram	2x1=2	
	(ii) (A) Acetic acid (or) formula	½+½ =1	5
	(B) Arsenic (or) formula	2x1=2	
36 (a)	(i) Effect of catalyst on reaction rate:		
	Energy of activation is lowered	1	
	Hence greater number of molecules can cross the energy Barrier and change over to products. (or)	2	5
	Diagram only	1	
	(ii) Lewis acids: BF_3, CO_2 Lewis bases: MgO, CH_3^-	4x½=2	

... are stored in the presence of atmospheric oxygen, they slowly oxidise to form hydroperoxides and dialkyl peroxides.



(b) (i) <u>Baeyer's reagent:</u>

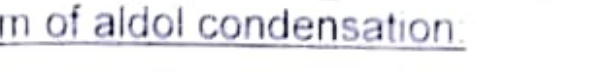
Cold alkaline $KMnO_4$



(ii) Antiseptic – Applied to living tissue
Disinfectant – Generally used on inanimate objects

(i) Mechanism of aldol condensation:

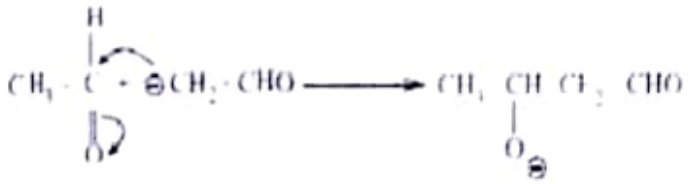
The carbanion is formed as the α-hydrogen atom is removed as a proton by a base



1½

Step 2

The carbanion attacks the carbonyl carbon of another unionized aldehyde to form an alkoxide ion

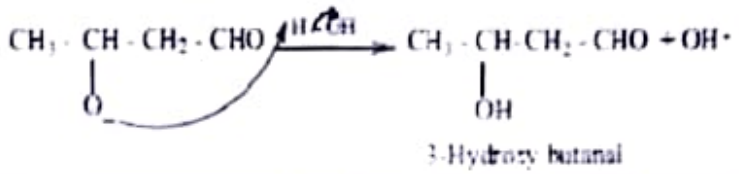


1½

5

Step 3

The alkoxide ion formed is protonated by water to form aldol



3-Hydroxy butanal

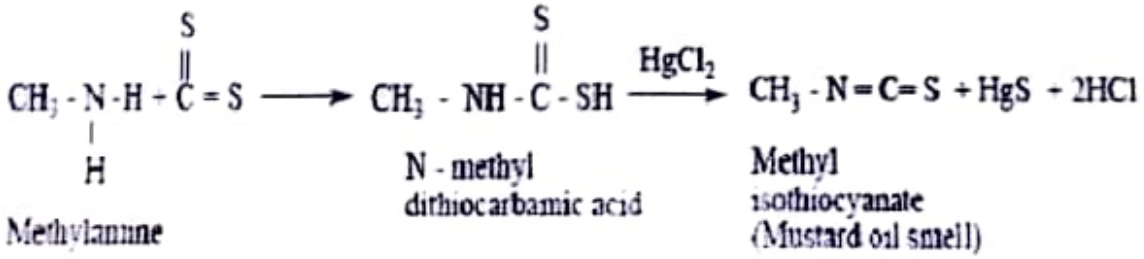
2

(i) Reducing agent used in the reduction of nitrobenzene:

(A) Sn/HCl
(B) Zn/NH_4Cl
(C) Fe/H_2O (steam)

3x1=3

(ii) Mustard oil reaction:



Methylamine — N - methyl dithiocarbamic acid — Methyl isothiocyanate (Mustard oil smell)

2x1=2

5

(OR)
Explanation only

1

DEPARTMENT OF GOVERNMENT SUPLEMENTARY EXAMINATIONS
HIGHER SECONDARY YEAR EXAMINATIONS AUGUST - 2021
CHEMISTRY ANSWER KEY (NEW SYLLABUS)

Note:

1. Answer written with Blue or Black Ink only to be evaluated.
2. Choose the most suitable answer in part A from the given alternatives and write the option code and the corresponding answer.
3. For answers in Part-II, Part-III and Part IV like reasoning, explanation, narration, description and listing of points, students may write in their own words but without changing the concepts and without skipping any point
4. In numerical problems, if formula is not written, marks should be given for the remaining correct steps.
5. In graphical representation, physical variables for X-axis and Y-axis should be marked.

PART- (I)

Each Correct answer carries one mark 15 X1=15

TYPE - A			TYPE -B		
S.No.	Option	Answer	S.No.	Option	Answer
1.	c)	Tyndall effect	1.	d)	FeO
2.	d)	Van Arkel Process	2.	a)	Cytosine and Thiamine
3.	d)	1-iii, 2-iv, 3-i, 4-ii	3.	d)	Strongly Acidic
4.	a)	Cytosine and Thiamine	4.	d)	Violet Colour
5.	d)	Strongly Acidic	5.	d)	Victor Meyer test
6.	d)	Violet Colour	6.	c)	Acetic acid
7.	d)	Hexane-1, 6-diamine	7.	a)	$6.022X10^{22}$
8.	a)	$6.022X10^{22}$	8.	d)	1-iii, 2-iv, 3-i, 4-ii
9.	c)	Diamond	9.	b)	diagram
10.	c)	Acetic acid	10.	c)	NeF_2
11.	d)	FeO	11.	c)	Tyndall effect
12.	d)	Victor Meyer test	12.	d)	6
13.	b)	diagram	13.	d)	Van Arkel Process
14.	c)	NeF_2	14.	c)	Diamond
15.	d)	6	15.	d)	Hexane-1, 6-diamine

PART II

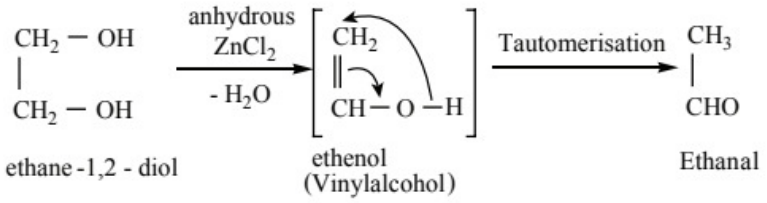
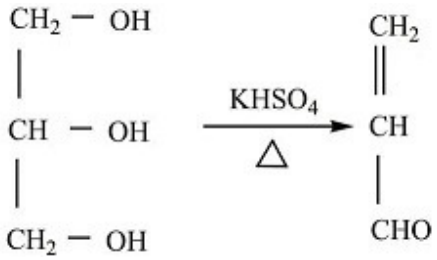
Answer any six questions. 6X2 = 12
Q.No 24 is compulsory

16	Any Two Uses	2 X1	2
17	The Energy difference between (n-1)d and ns orbital's are very small		2
18	Correct Definition		2
19	Correct law (or) $\propto = \sqrt{\frac{Ka}{C}}$ (or) $\propto = \sqrt{\frac{Kb}{C}}$		2
20	Correct definition (or) correct formula		2
21	Any two factors	2 X 1	2
22	Movement of dispersion medium under the influence of electric Potential		2
23	Correct Explanation (or) Peptide bond equation		2
24	(A) $CH_3 – NH_2$ (or) Methylamine (or) Methanamine (or) Aminomethane (B) $CH_3 – NHOH$ (or) N- Methyl hydroxylamine	1 1	2

PART III

Answer any six questions : Q.No 33 is compulsory 6X3 = 18

25	Correct Explanation Coordination Number 8	2 1	3
26	Each halogen combines with other halogens to form a series of compounds Any two examples	2 2X ½ 1	3
27	Two differences	2X 1½	3

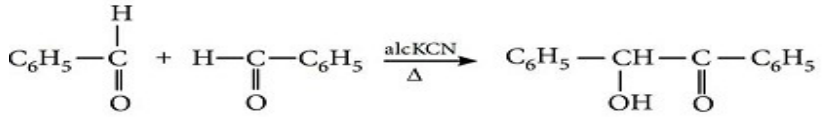
28	1. Small Size of the first member 2. High I.E and High EN 3. Absence of d orbitals in their valence shell	1 1 1	3
29	I law definition II law definition	1½ 1 ½	3
30 (i)	 Equation without catalyst - 1 Marks Explanation only - 1 Marks	1 ½	3
30 (ii)	 Catalyst: $KHSO_4$ (or) Conc. H_2SO_4 Equation without catalyst - 1 Marks Explanation only - 1 Marks	1 ½	
31	Three tests	3 X1	3
32	Any three differences	3 X1	3
33	<u>Covalent Crystals</u> Diamond , SiO_2 <u>Molecular Crystals</u> Naphthalene , glucose <u>Ionic Crystal</u> NaCl <u>Metallic Crystal</u> Brass	2 X ½ 2 X ½ 1 X ½ 1 X ½	3

Part – IV

Answer all questions: 5 X 5 = 25

34 (a)	Sulphide ores are concentrated by froth floatation process Water + Pine oil + Eucalyptus oil + Sodium ethyl Xanthate Foam is produced by Passing air into the mixture ore particles are wetted by the oil rise to the surface along with the froth Diagram (or)	1 1 1 1 1	5
34 (b)	i) Reducing property $SO_2 + 2H_2O \rightarrow H_2SO_4 + 2(H)$ $X+2(H) \rightarrow XH_2$ Coloured Colourless ii) Any two uses	1 1 1 2X1	5
35 (a)	i) An interstitial compound or alloy is a compound that is formed when small atoms like hydrogen, boron, carbon or nitrogen are trapped in the interstitial holes in a metal lattice ii) Ti^{+3} - 1Unpaired e^- Magnetic Moment $\mu = \sqrt{n(n+2)}\ \mu_B$ Mn^{+2} - 5 unpaired e^- Magnetic moment of Ti^{+3} = 1.73 μ_B Magnetic moment of Mn^{+2} = 5.91 μ_B (or)	2 ½ 1 ½ ½ ½	5

35 (b)	i) Three limitations ii) Electronic Configuration of Ni^{2+} ion CN^- - Strong field Ligand Paring of 3d electrons No of unparied e^- = 0	3X1 ½ ½ ½ ½	5
36 (a)	i) Any Two difference ii) $\text{Rate} = k\,[A]^0$ $\frac{-d[A]}{dt} = k\,(1) \qquad (\therefore [A]^0 = 1)$ $\Rightarrow -d[A] = k\,dt$ $-\int_{[A_0]}^{[A]} d[A] = k\int_0^t dt$ $-([A])_{[A_0]}^{[A]} = k\,(t)_0^t$ $k = \frac{[A_0]-[A]}{t}$ (or)	2X1 ½ ½ ½ ½ ½ ½	5
36 (b)	$pH = pK_a + \log\frac{[salt]}{[acid]}$ Given that $K_a = 1.8 \times 10^{-5}$ $\therefore pK_a = -\log(1.8\times10^{-5}) = 5 - \log 1.8$ $= 5 - 0.26$ $= 4.74$ $\therefore pH = 4.74 + \log\frac{0.20}{0.18}$ $= 4.74 + \log 10/9 = 4.74 + \log 10 - \log 9$ $= 4.74 + 1 - 0.95 = 5.74 - 0.95$ $= 4.79$	1 1 1 1 1	5

37 (a)	i)  Equation without Catalyst - 1 Marks Explanation only - 1 Marks ii) Knoevenagal reaction (or) Perkin's reaction Explanation only - 1 Marks iii) Equation with Conc H_2SO_4 Equation without Conc H_2SO_4 - 1 Marks Explanation only - 1 Marks (or)	1 ½ 2 1 ½	5
37 (b)	i) Lucas test for primary alcohol Lucas test for Secondary alcohol Lucas test for Tertiary alcohol Lucas test Explanation only - 2 Marks	1 1 1	5
	ii) Four uses - 4 X ½	2	
38 (a)	Five steps (or)	5 X 1	5
38 (b)	$CH_3-CN \xrightarrow[4[H]]{Na(Hg)/C_2H_5OH} CH_3CH_2-NH_2$ $C_2H_5-NH_2 + HNO_2 \xrightarrow[\text{Conc. HCl}]{NaNO_2} [C_2H_5\overset{+}{N_2}\overset{-}{Cl}] \xrightarrow{H_2O} C_2H_5OH + N_2$ A – CH_3– CN ----- Methyl Cyanide B - $CH_3CH_2NH_2$ ------- Ethylamine C - C_2H_5-OH -------- Ethyl alcohol or Ethanol	1 1 1 1 1	5

******

Answer Key Preparation Teachers Name & Address

S.No	Name	School Name	Phone No.
1.	K.Selvakumar PG Teacher Chermistry	Municipal Higher Secondary School, Melagandhinagar Sattur – 626203.	9442380101
2.	K.Balamurugan PG Teacher Chemistry	Saliary Mahajana Higher Secondary School, Aruppukottai – 626 101	9786397614

DIRECTORATE OF GOVERNMENT EXAMINATION, CHENNAI – 6
HIGHER SECONDARY SECOND YEAR EXAMINATION – MAY 2022
CHEMISTRY KEY ANSWER

NOTE:

1. Answers written with **Blue or Black** ink only to be evaluated.
2. PART – I, choose the correct answer and write with option code. If one of them (option or answer) is wrong, then award zero mark only.

MAXIMUM MARKS : 70

PART - I

15 x 1 = 15

TYPE - A			TYPE - B		
Q.No	**OPTION**	**ANSWER**	**Q.No**	**OPTION**	**ANSWER**
1	**c**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	1	**c**	Glycine
2	**a**	$Al_2O_3\ .nH_2O$	2	**a**	Methanal
3	**d**	30 minutes	3	**a**	basic
4	**c**	Sn/HCl	4	**d**	FeO
5	**a**	Methanal	5	**c**	Hydrolysis of sucrose in presence of dil.HCl
6	**c**	$[Fe(CO)_5]$	6	**c**	Sn/HCl
7	**a**	HPO_4^{2-}	7	**d**	charge carried by one mole of electron
8	**a**	basic	8	**a**	$Al_2O_3\ .nH_2O$
9	**c**	Hydrolysis of sucrose in presence of dil.HCl	9	**d**	30 minutes
10	**c**	nucleophilic addition	10	**a**	HPO_4^{2-}
11	**c**	+3	11	**c**	$[Fe(CO)_5]$
12	**d**	charge carried by one mole of electron	12	**c**	+3
13	**c**	Glycine	13	**c**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
14	**d**	FeO	14	**a**	liquid in gas
15	**a**	liquid in gas	15	**c**	nucleophilic addition

PART – II

Q.No 24 is compulsory **6 x 2 = 12**

Q.No	Answer	Marks	
16	**Difference between minerals and ores** Any two differences	2 x1	**2**
17	**Stability of Fe^{2+} & Fe^{3+}** Fe^{3+} is more stable Half filled d orbital **(or)** d^5 configuration **(or)** Electronic configuration of Fe^{3+}	1 1	**2**
18	**Coordination number** Any one correct definition		**2**
19	**Covalent solids** Correct definition **(or)** entirely by covalent bonds	2 1	**2**
20	**Examples of First order reaction** Any two examples Equation (or) Explanation	2 x 1	**2**
21	**Limitation of Arrhenius concept** Two Limitations	2 x 1	**2**
22	**Electrophoresis** Correct definition	2	**2**
23	**IUPAC Names** a. 2-methyl propan-2-ol **(or)** 2- methyl -2-propanol b. phenyl methanol	1 1	**2**
24	**Identificaton of A & B compounds** A = CH_3 - N_3 (or) Methyl azide B = CH_3 - NH_2 (or) Methylamine	1 1	**2**

PART – III

Q.No 33 is compulsory **6 x 3 = 18**

Q.No	Answer	Marks	
25	**Interhalogen compounds** Correct definition Any one example	 2 1	**3**
26	**Properties of interstitial compound** Any three properties	3 x 1	**3**
27	**Arrhenius equation** $k = Ae^{-\left(\frac{E_a}{RT}\right)}$ Four terms explanation	1 4 x ½	**3**
28	**Factors that affects electrolytic conductance** • Dielectric constant of Solvent • Viscosity of the medium • Temperature of the solution • Dilution of the solution (Any three factors)	3 x 1	**3**
29	**Homogeneous catalysis** Correct definition Any one example (Equation (or) Explanation)	2 1	**3**
30	**Preparation of diethyl ether** Any one equation of preparation with suitable condition. **(or)** Equation without condition (or) Explanation only	3 2	**3**
31	**Haloform reaction** Any one correct equation **(or)** Explanation only	3 2	**3**

Q.No	Answer	Marks	
32	**Epimers** Correct explanation Any one example	 2 1	**3**
33	**$[Ag(NH_3)_2]^+$ complex** Ligand : NH_3 (or) ammine Central metal ion : Ag^+ (or) silver (I) ion (or) Ag (I) IUPAC name : diamminesilver(I) ion	 1 1 1	**3**

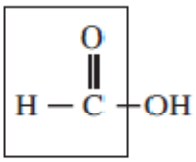
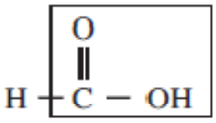
PART – IV

5 x 5 = 25

Answer all the questions

34 a	(i) **Gravity separation method** Correct explanation **(or)** Any one example of ores	 2 1	**5**
	(ii) **Mond's process** Two Correct equations with temperature **(or)** Equation without temperature (or) explanation only	 3 2	
34 b	(i) **Inert pair effect** Correct definition	 2	**5**
	(ii) **Uses of boric acid** Any three uses	 3 x1	
35 a	(i) **Uses of Oxygen** Any two uses	 2 x1	**5**
	(ii) P**reparation of bleaching powder** Correct equation **(or)** Explanation only	 3 2	

Q.No	Answer	Marks	
35 b	**Postulates of Werner's theory** Five Postulates	5 x1	**5**
36 a	**Crystalline and amorphous solids differences** Any Five differences	5 x1	**5**
36 b	**(i) pH definition** Correct definition (or) pH = $-\log_{10}[H_3O^+]$ (or) pH = $-\log_{10}[H^+]$	2	**5**
	(ii) Common ion effect Correct definition Any one example (Equation (or) Explanation)	2 1	
37 a	**Nernst equation** $xA + yB \rightleftharpoons lC + mD$ $Q = \frac{[C]^l [D]^m}{[A]^x [B]^y}$ $\Delta G = \Delta G^\circ + RT \ln Q$ $\Delta G = -nFE_{cell}$; $\Delta G^\circ = -nFE^\circ_{cell}$ $-nFE_{cell} = -nFE^\circ_{cell} + RT \ln \frac{[C]^l [D]^m}{[A]^x [B]^y}$ $E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \ln \frac{[C]^l [D]^m}{[A]^x [B]^y}$ (or) $E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF} \log \frac{[C]^l [D]^m}{[A]^x [B]^y}$ (or) $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[C]l\,[D]m}{[A]x[B]y}$	½ ½ 1 ½ +½ 1 1	**5**
37 b	**Characteristics of catalyst** Any Five characteristics	5 x 1	**5**

Q.No	Answer	Marks	
38 a	**Reducing action of formic acid** Contains both an aldehyde as well as an acid group **(or)**  Aldehyde group  Carboxylic acid group Any one example with correct equation **(or)** Explanation only	2 3 2	**5**
38 b	(i) **Carbylamine reaction** Correct equation **(or)** Explanation only. **(ii) Gabriel phthalimide synthesis** Correct equation **(or)** Explanation only.	2 1 3 2	**5**

DEPARTMENT OF GOVERNMENT EXAMINATIONS – CHENNAI-6

HSC SECOND YEAR EXAMINATION MARCH/APRIL - 2023

CHEMISTRY ANSWER KEY

Note: 1. Answer written with Blue or Black ink only to be evaluated
2. Choose the most suitable answer in **PART – I** from the given alternatives and write the option code and the corresponding answer.

Maximum Marks : 70

PART – I

Answer all the questions **15×1=15**

Q.No	Option	'A' Type	Q.No	Option	'B'Type
1	c)	Antacid	1	a)	NaCl
2	c)	Activation energy	2	a)	Uracil
3	b)	Al	3	c)	Antacid
4	c)	$[Cu(NH_3)_4]^{2+}$	4	d)	+3
5	d)	Gel-butter	5	b)	Ethane – 1,2-diol
6	d)	HCl	6	a)	sp^2
7	a)	sp^2	7	b)	Al
8	d)	+3	8	a)	Schiff's base
9	d)	Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**	9	b)	0
10	c)	Rn	10	c)	$[Cu(NH_3)_4]^{2+}$
11	b)	0	11	c)	Activation energy
12	a)	NaCl	12	d)	HCl
13	a)	Uracil	13	d)	Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of
14	b)	Ethane – 1,2-diol	14	c)	Rn
15	a)	Schiff's base	15	d)	Gel-butter

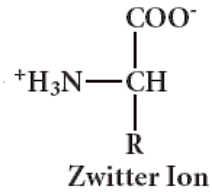
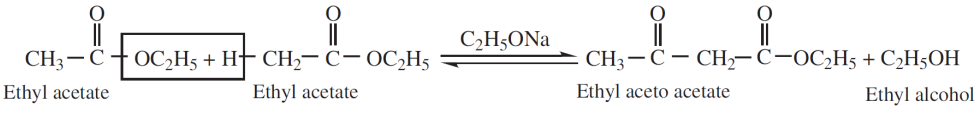
Part –II

Answer any **SIX** Questions and **Question No.24** is Compulsory. **6×2=12**

16	**Sulphide ore** **Galena, Zinc blende (or) any two suitable examples with name or formula**	1 ½+½	2
17	Any two uses	1+1	2
18	**Central atom** Correct Definition		2
19	Number of atoms in FCC unit cell = **Nc / 8 + N_f / 2** **(or)** **= 8/8 + 6/2** **= 4** (or) Correct Structure	1 1 1	2
20	**Conjugate acid – base pairs** Chemical species that differ only by a proton (or) suitable explanation (or) mentioning any one conjugate acid base pair	2 1	2
21	correct explanation	2	2
22	$CH_3-\mathbf{CO}-CH_3 + 4[H] \xrightarrow{Zn/Hg\ /\ Con\ HCl} CH_3-\mathbf{CH_2}-CH_3$ Acetone Propane (OR) $CH_3-\mathbf{CO}-CH_3 + 4[H] \xrightarrow{NH_2-NH_2\ /\ C_2H_5ONa} CH_3-\mathbf{CH_2}-CH_3$ Acetone Propane (or) Correct explanation.	2 1	2
23	Correct explanation Any one example	1 1	2
24	(A) - $CH_3CH_2NH_2$ (or) Ethyl amine (or) ethanamine (B) - $CH_3CH_2NHCOCH_3$ (or) N-ethylacetamide	1 1	2

Part–III

Answer any **SIX** Questions and **Question No.33 is Compulsory.** 6×3=18

25	**Fisher tropsch synthesis:** $nCO + (2n+1)H_2 \xrightarrow{500-700K, \text{less than } 50 \text{ atm}} C_nH_{2n+2} + nH_2O$ (or) $nCO + 2nH_2 \xrightarrow{500-700K, \text{less than } 50 \text{ atm}} C_nH_{2n} + nH_2O$ (or) unbalanced equation(or) equation without condition (or) mere explanation alone	3 2	3
26	Any three differences	3×1	3
27	a) Central metal atom / ion = Pt (or) Pt^{2+} (or) Pt(II) b) Co-ordination number = 4 c) Oxidation number of central metal ion= +2	1 1 1	3
28	**Helmholtz electrical double layer:** Correct explanation **(or)** Diagram alone	3 2	3
29	**First Law : Correct statement** **(or) Correct mathematical expression** **Second Law : Correct statement** **(or) Correct mathematical expression**	1½ 1 1½ 1	3
30	 Zwitter Ion (or) any other correct structure		3
31	 Ethyl acetate Ethyl acetate Ethyl aceto acetate Ethyl alcohol (or) equation without sodium ethoxide (or) **Mere** Explanation	3 2 ½ 2	3

32	Correct explanation Any two Examples	2 ½+½	 3
33	$t = \frac{2.303}{k} \log \frac{[A_0]}{[A]}$ $t_{90\%} = \frac{2.303}{k} \log \frac{[100]}{[100-90]}$ (or) $t_{90\%} = \frac{2.303}{k} \log(10)$ $t_{99\%} = \frac{2.303}{k} \log \frac{[100]}{[100-99]}$ (or) $t_{99\%} = \frac{2.303}{k} \log(100)$ $t_{99\%}/t_{90\%} = \frac{\log 100}{\log 10}$ (or) $t_{99\%}/t_{90\%} = 2$	1 ½ ½ 1	3

Part– IV

Answer all the Questions **5x5=25**

34 (a)	**Zone Refining** Principle - fractional crystallization Correct explanation Example: Germanium (Ge) / silicon (Si) / gallium (Ga) /Semiconductor **(OR)**	 1 3 1	5
(b)	(i). (1) - 1 (2) + 4 (ii). (1) $P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3 \uparrow$ (2) $XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$ (3) $Cu + 2H_2SO_4 \longrightarrow CuSO_4 + 2H_2O + SO_2 \uparrow$ con. (or) Unbalanced equations - ½ + ½ +½	1+1 **1** **1** 1 **1½**	2 3

35 (a)	(i). Correct Structure **Any four points from the following.** 1. Two BH_2 units are linked by two bridged hydrogens. 2. It has eight B-H bonds. 3. It has only 12 valence electrons unable to form normal covalent bonds. 4. The four terminal B-H bonds (2c-2e) bond. 5. Two B-H-B (3c-2e) or bridged bond. 6. The bridging hydrogen atoms are in a plane 7. The boron is sp^3 hybridized.	1 4×½	**3**
	(ii). **Ethyl Borate test** $H_3BO_3 + 3C_2H_5OH \xrightarrow[H_2SO_4]{Conc.} B(OC_2H_5)_3 + 3H_2O$ (or) Equation without conc.sulphuric acid (or) correct explanation (or) mentioning triethyl borate or green flame (or) unbalanced equation **(OR)**	2 1½ 1	2
(b)	**Bonding in metal carbonyls** i. The bond between metal atom and the carbonyl ligand consists of two components. ii. $M \xleftarrow{\sigma\ bond} CO$ **sigma bond**. (or) explanation ii. The sigma bond formation increases the electron density in metal d orbitals. iv. Correct explanation for **π-back bonding** (or) suitable diagram	1 1 1 2	5
36 (a)	**Schottky defect:** Correct reason Similar size (or) density decreases. Example: NaCl. Diagram **Frenkel defect:** Correct reason differ in size (or) does not affect the density Example: AgBr Diagram **(OR)**	1 ½ ½ ½ 1 ½ ½ ½	5

36 (b)	(i) Any two correct examples for a zero order reaction (ii) **uses of colloids** in Tanning of leather (one use) in Rubber industry (one use)	2 1½ 1½	2 3
37 (a)	Oswald dilution law $CH_3COOH \leftrightharpoons H^+ + CH_3COO^-$ $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$ $K_a = \frac{(\alpha.C)(\alpha.C)}{(1-\alpha)C}$ $k_a = \frac{\alpha^2 C}{1-\alpha}$ $\alpha = \sqrt{\frac{K_a}{C}}$ (or) $[H^+] = \sqrt{K_a C}$ (OR)	1 1 1 1 1	5
(b)	(i). Aniline is basic in nature It donates its lone pair to the lewis acid to form an adduct / inhibits further the electrophilic substitution reaction. (ii). Correct equation (or)Mere explanation alone	1 1 3 2	2 3
38 (a)	(i). Correct equation Correct equation without conc.sulphuric acid (or) Mere explanation alone (ii) . Correct equation Correct equation without Na / ether (or) Mere explanation only (OR)	3 2½ 2 2 1½ 1	3 2
(b)	$C_6H_5-OH + NH_3 \xrightarrow[\Delta]{anhy.ZnCl_2} C_6H_5-NH_2$ (A) (B) $C_6H_5-OH + Zn \xrightarrow{\Delta} C_6H_6 + ZnO$ (C) (A) -C_6H_5OH (or) Phenol (B) -$C_6H_5NH_2$ (or) Aniline (C) - C_6H_6 (or) Benzene	1 1 1 1 1	5

Directorate of Government Examination, Chennai - 6
Higher Secondary Second Year Examination – June 2023
CHEMISTRY- KEY

Note:

1) Answer written with Blue or Black ink only to be evaluated.
2) **Part-I**, Choose the Correct answer and write the option code, if one of them (option or Answer) is wrong then award zero mark only.

Maximum Marks: 70

Part -I **15x1=15**

Q.No	Option	Answer
1	C	(1)-(iv), (2)-(iii), (3)-(ii), (4)-(i)
2	C	four
3	A	H_3PO_3
4	B	carbon dioxide
5	A	3
6	C	both covalent crystals
7	A	First order
8	B	PF_3
9	C	6.022×10^{22}
10	D	ΔS decreases
11	A	Phenol
12	B	Victor Meyer test
13	D	Yellow colour solution
14	D	Vitamin B
15	A	Acetylsalicylic acid

Part – II

Answer any SIX Questions and Questions No.24 is Compulsory. **6 X 2 =12**

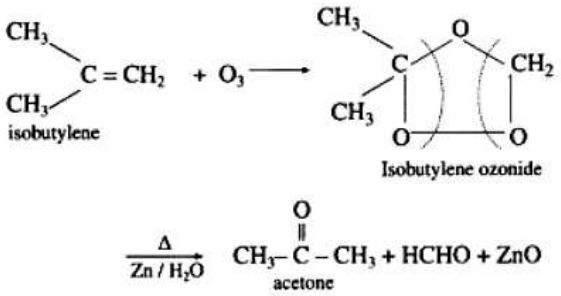
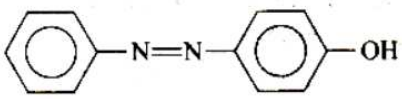
16	(i) Icosagen – any one example (ii) Chalcogen - any one example	1 1	2

17	Any one balanced equation (or) unbalanced equation (or) mere explanation alone	2 1½ 1	2
18	Correct Explanation	2	2
19	Correct Definition	2	2
20	Any two characteristics	2x1	2
21	1,4-Dioxane (or) correct structure	2	2
22	Correct Definition Any one use	1 1	2
23	Correct Explanation Any two examples	1 ½+½	2
24	$K_w = [H_3O^+][OH^-]$ $\therefore [OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1\times10^{-14}}{2\times10^{-3}} = 5\times10^{-12} M$ (or) the juice is acidic in nature	1 ½+½ 1	3

Part – III

Answer any SIX Questions and Questions No.33 is Compulsory. 6 X 3 =18

25	Two limitations	1½ + 1½	3
26	Any three uses	3x1	3
27	Oxidation number of central metal ion: +2 Co-ordination number: 6 Nature of ligand: strong field ligand	1 1 1	3
28	Any three differences	3x1	3
29	Correct reason	3	3
30	**M_1** is better for coating the surface of iron. **Reason:** The oxidation potential of **M_1** is more than the oxidation potential of iron (or) any suitable explanation with reason	1 2 2	3

31	 (or) gives formaldehyde and acetone	1½ + 1 ½ 1+1	3
32	Correct explanation	3	3
33	A – $C_6H_5NH_2$ (or) Aniline B – $C_6H_5N_2Cl$ (or) Benzene diazonium chloride C -  (or) p-hydroxy azo benzene	1 1 1	3

Part – IV

Answer all the Questions **5 x 5 = 25**

34 (a)	**(i) Refining of Nickel** **$Ni_{(s)}$ + 4 CO $_{(g)}$ $\xrightarrow{350\,K}$ $[Ni(CO)_4]_{(g)}$** **$[Ni(CO)_4]_{(g)}$ $\xrightarrow{460\,K}$ $Ni_{(s)}$ + 4 CO $_{(g)}$** (or) Equation without temperature (or) mere explanations **(ii) liquation** Correct Explanation **(OR)**	1½ 1½ (1+1) 2	**5**
(b)	**(i) Hydroboration** $B_2H_6 + 6RCH{=}CHR \longrightarrow 2(RCH_2\text{-}CHR)_3B$ (or)Unbalanced equation (or) mere explanation (or) mentioning anti Markovnikov addition **(ii) Bleaching action of SO_2** Correct Explanation (or) equation	3 2 1 2	**5**
35. (a)	**Lanthanoid Contraction** Correct Explanation **Effects of Lanthanoid Contraction** Any three effects **(OR)**	2 3x1	5

(b)	**Structure of three geometrical isomers** two cis and one trans isomers - **Structures** **Optically active isomers:** mentioning **two** cis isomers	1+1+1 2	5
36 (a)	**Crystalline solids and amorphous solids** Any five differences	5x1	5
	(OR)		
(b)	**Zero order reaction** Rate = k $[A]^0$ $\frac{-d[A]}{dt} = k\ (1)$ (or) $-d[A] = kdt$ $-\int_{[A_0]}^{[A]} d[A] = k\int_0^t dt$ $-([A])_{[A_0]}^{[A]} = k(t)_0^t$ $[A_0] - [A] = kt$ $k = \frac{[A_0]-[A]}{t}$	1 1 1 ½ ½ 1	5
37 (a)	**The intermediate compound formation theory** Homogeneous catalysis Theory - explanation any suitable example with intermediate compound formation	1 2 2	5
	(OR)		
(b)	**Importance of proteins** Any five points	5x1	
38 (a)	**Lucas test** $1^0, 2^0, 3^0$ alcohol equations with conditions explanation - turbidity formation for $1^0, 2^0, 3^0$ alcohols (or) mentioning con.HCl and anhy.$ZnCl_2$	1+1+1 2 1	
	(OR)		
(b)	**Schotten Baumann reaction** Correct equation (or) without mentioning the catalyst (or) mere explanation alone **Mustard oil reaction** Correct equation (or) without mentioning the catalyst (or) mere explanation alone	2½ 2 1 2½ 2 1	5

DEPARTMENT OF GOVERNMENT EXAMINATIONS – CHENNAI-6
HSC SECOND YEAR EXAMINATION MARCH - 2024
CHEMISTRY ANSWER KEY

Note: 1. Answer written with Blue or Black ink only to be evaluated

2. Choose the most suitable answer in **PART – I** from the given alternatives and write the option code and the corresponding answer.

Maximum Marks: 70

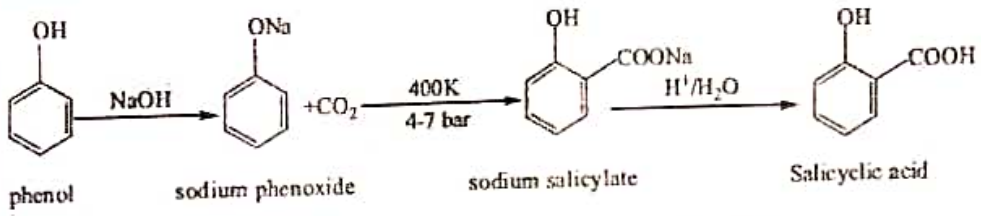
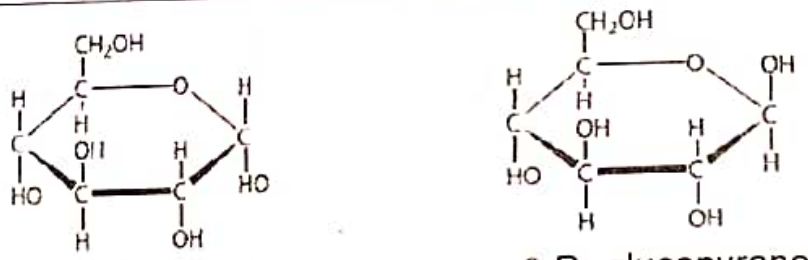
PART- I

Answer all the questions **15×1=15**

Q.No	Option Code	'A' Type	Q. No	Option Code	'B' Type
1	b)	Both Assertion and Reason are true and Reason is the correct explanation of Assertion	1	b)	First order
2	c)	Potassium trioxalato aluminate (III)	2	b)	Acetyl salicylic acid
3	b)	HI	3	d)	carbon dioxide
4	c)	Dry ice	4	c)	Potassium trioxalato aluminate (III)
5	c)	Cytosine and Uracil	5	a)	Sodium chloride
6	b)	Acetyl salicylic acid	6	b)	(i) and (iv)
7	d)	carbon dioxide	7	d)	Impure copper
8	c)	acetanilide	8	b)	Both Assertion and Reason are true and Reason is the correct explanation of Assertion
9	a)	Sodium chloride	9	c)	Nucleophilic addition
10	b)	(i) and (iv)	10	c)	Dry ice
11	c)	Nucleophilic addition	11	d)	PCC
12	b)	First order	12	b)	HI
13	d)	Impure copper	13	c)	acetanilide
14	c)	Al_2O_3	14	c)	Cytosine and Uracil
15	d)	PCC	15	c)	Al_2O_3

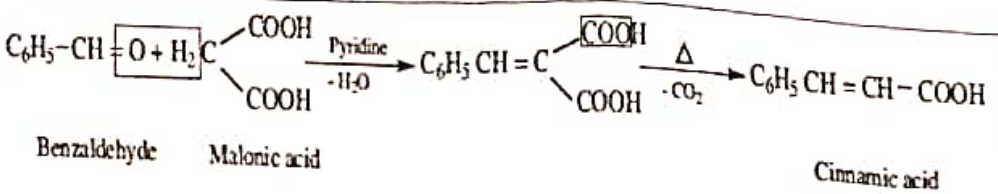
Part –II

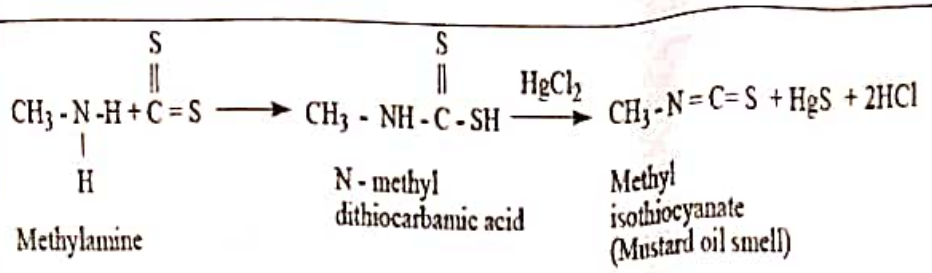
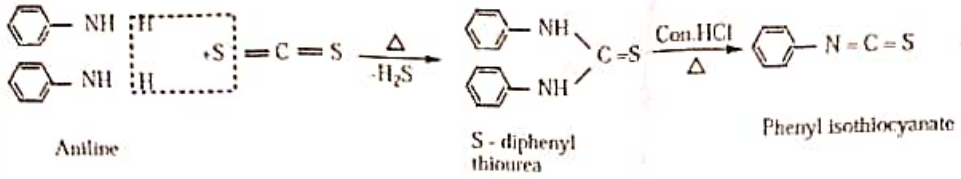
Answer any SIX Questions and Question **No.24 is Compulsory.** **6×2=12**

Q.No	Answer	Marks	
16	Correct Explanation **(or)** Correct Equation	2	2
17	Correct equation **(or)** unbalanced equation **(or)** equation without condition **(or)** mere explanation alone	2 1½ 1	2
18	Any one correct equation **(or)** Explanation (Any one)	2 1	2
19	Correct explanation Example	1 1	2
20	$E^{\circ}_{cell} = (E^{\circ}_{oxi}) + (E^{\circ}_{red})$ **(or)** $= -1.09+0.771$ $E^{\circ}_{cell} = -0.319$ V **(or)** E°_{cell} is –ve , Fe^{3+} cannot oxidises Br^- to Br_2	½ ½ 1	2
21	 **(or)** equation without condition **(or)** explanation only	2 1	2
22	 α-D- glucopyranose β-D- glucopyranose	1+1	2
23	Correct explanation **(or)** one example	2 1	2
24	correct definition	2	2

Part–III

Answer any **SIX** Questions and **Question No.33 is Compulsory.** **6×3=18**

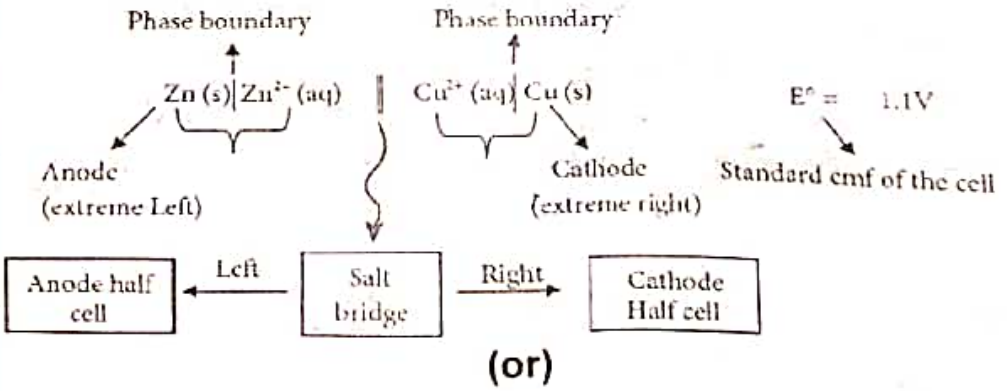
Q.No	Answer	Marks	
25	Any three uses	3	3
26	Fe^{3+}= [Ar] $3d^5$ Fe^{3+} - half-filled d orbital Fe^{3+} is stable	1 1 1	3
27	$r = \frac{a\sqrt{2}}{4}$ **(or)** $a = \frac{4r}{\sqrt{2}}$ **(or)** $a = 2\sqrt{2}\,r$ **(or)** a=2x1.414 r $a = \frac{4 \times 125}{\sqrt{2}}$ **(or)** $a = 2\sqrt{2}\,125$ **(or)** a=2x1.414 x125 a = 353.5 pm	1 1 ½+½	3
28	$K = Ae^{-Ea/RT}$ K = Rate constant A = Frequency factor R = Gas constant E_a = Energy of activation T =Temperature	1 4x½	3
29	**Physisorption** / **chemisorption** Physisorption decreases with increase in temperature. / When temperature is raised chemisorption first increases and then decreases. In Physisorption, when pressure increases the extent of adsorption increases. / Chemical adsorption is fast with increase Pressure, it cannot alter the amount.	1½ 1½	3
30	 **(or)** equation without condition **(or)** explanation only	3 2½ 1	3

31	 (or) 	3	3
	(or) equation without condition	2½	
	(or) explanation only	1	
32	Correct explanation **(or)** correct example	3	3
	(or) -CO-NH- bond only	1	
33	Mere Attempt	3	3

Part– IV

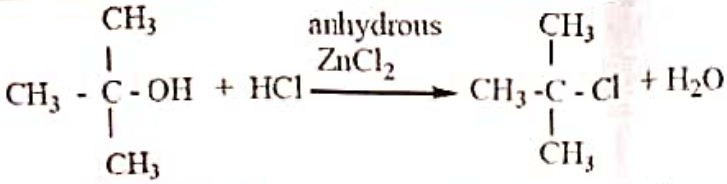
Answer all the Questions **5x5=25**

Q.No	Answer	Marks	
34 (a)	i) Any 3 differences	3	5
	ii) Silica acts as a flux **(or)** Correct Equation only	2	
	(OR)		
(b)	i) Any 3 uses	3	5
	ii) Correct definition	2	
	(or) Correct Structure only	1	
35 (a)	Lanthanoid contraction – Explanation	2	5
	Lanthanoid contraction consequences **(Any Three)**	3	
	(OR)		
(b)	i) Double salt – Explanation **(or)**	1½	5
	Double salt – Example	1	
	Co-ordination Compound – Explanation **(or)**	1½	
	Co-ordination Compound – Example	1	
	ii) One example for medicinal importance	1	
	One example for biological importance	1	

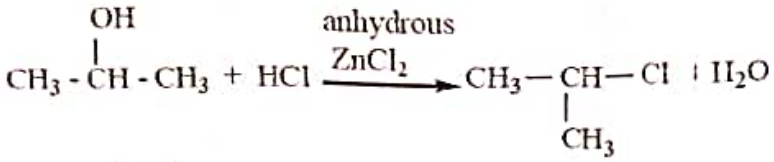
36 (a)	Packing efficiency = $\frac{\text{total volume occupied by spheres in a unit cell}}{\text{volume of the unit cell}} \times 100$	1	
	Volume of cube = a x a x a = a^3	1	
	(or)		
	Radius of the sphere from figure, a = 2r(or)r = $\frac{a}{2}$		
	Volume of the sphere with radius = $\frac{4}{3}\pi\left(\frac{a}{2}\right)^3$	1	5
	(or) $= \frac{\pi a^3}{6}$		
	Packing efficiency = $\frac{1 \times \frac{\pi a^3}{6}}{a^3} \times 100$	1	
	= 52.38%	1	
	(OR)		
(b)	i) Rate = $k[A]^0$ (or) $\frac{-d[A]}{dt} = k$ (or) - d[A]= kdt	1	
	$-\int_{[A_0]}^{[A]} d[A] = K\int_0^t dt$ (or) $-([A])_{[A_0]}^{[A]} = k\,(t)_0^t$	1	5
	$[A_0] - [A] = kt$ (or) $k = \frac{[A_0]-[A]}{t}$	1	
	ii) Correct definition (or)	2	
	Formula	1	
37 (a)	i) 	3	5
	(or)		
	Correct Explanation	3	
	(or)		
	$Zn \mid Zn^{2+} \mid\mid Cu^{2+} \mid Cu \quad E^0 = 1.1V$	1	
	ii) Correct definition	2	

(OR)

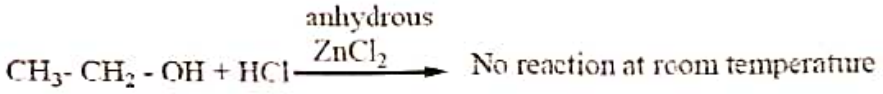
(b)



2-methylpropan-2-ol — 2 -chloro-2-methylpropane (immediate appearance of turbidity) — **2**



propan-2-ol — 2 -chloropropane (slow appearance of turbidity) — **2**



ethanol — (Turbidity appears only on heating) — **1**

(or) Correct equation without mentioning turbidity — **3**

(or) Correct equation without mentioning Lucas reagent — **3**

5

38 (a)

i)Any one correct equation — **2**

(or)

Correct Explanation — **1**

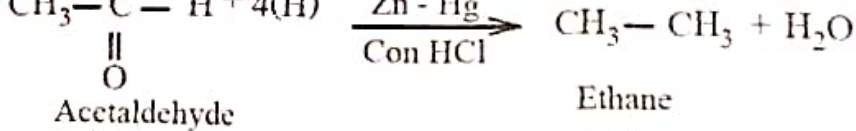
ii)Correct definition — **2**

Any one example — **1**

5

(OR)

(b)



Acetaldehyde **(A)** — Ethane **(B)** — **2**

Compound	Formula		Name
A	CH_3CHO	(or)	Acetaldehyde
B	CH_3CH_3		Ethane
E	CH_3COOH		Acetic acid

3×1

5